Principles of Artificial Intelligence

EX:1--N QUEENS PROBLEM

```
N = 8
count = 0

def print_board(board):
    for row in board:
        print(" ".join("Q" if cell == 1 else "." for cell in row))
    print()

def is_safe(board, row, col):
    for i in range(col):
        if board[row][i] == 1:
            return False

    i, j = row, col
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1
```

```python
    return True


def solve(board, col):
    global count


    if col == N:
        count += 1
        print(f"Solution {count}:")
        print_board(board)


        if count == 7:
            return True
        return False


    for row in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1


            if solve(board, col + 1):
                return True


            board[row][col] = 0


    return False


board = [[0] * N for _ in range(N)]
solve(board, 0)


print("Finished")
```

OUTPUT:

```
PS C:\Users\Lenovo> python 8queens.py
Solution 1:
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .

Solution 2:
Q . . . . . . .
. . . . . . Q .
. . . Q . . . .
. . . . . Q . .
. . . . . . . Q
. Q . . . . . .
. . . . Q . . .
. . Q . . . . .

Solution 3:
Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . . Q .
. . . Q . . . .
. Q . . . . . .
```

. . . . Q . . .

Solution 4:

Q . . . .

. . . . Q . . .

. . . . Q

. . . . . Q . .

. . Q . . . . .

. . . . . . Q .

. Q . . . . . .

. . . Q . . . .

Solution 5:

. . . . . Q . .

Q . . . .

. . . . Q . . .

. Q . . . . . .

. . . . Q

. . Q . . . . .

. . . . . . Q .

. . . Q . . . .

Solution 6:

. . . Q . . . .

Q . . . .

. . . . Q . . .

. . . . Q

. Q . . . . . .

. . . . . . Q .

. . Q . . . . .

. . . . . Q . .

Solution 7:

. . . . Q . . .

Q . . . .

. . . . Q

. . . Q . . . .

. Q . . . . . .

. . . . . . Q .

. . Q . . . . .

. . . . . Q . .

Finished